

Introduzione a YAML



Cos'è YAML

YAML, che sta per “**Y**AML **A**in't **M**arkup **L**anguage”, è un linguaggio per la rappresentazione delle informazioni.

Nasce come risposta, leggera e leggibile, a XML ed è molto utilizzato soprattutto per la costruzione di file di configurazione.

A differenza del XML, per definire delle strutture, non si utilizzano i tag annidati, bensì gli spazi dell'indentazione.

XML vs JSON vs YAML

```
<Servers>
  <Server>
    <name>Server1</name>
    <owner>John</owner>
    <created>12232012</created>
    <status>active</status>
  </Server>
</Servers>
```

XML

```
{
  Servers: [
    {
      name: Server1,
      owner: John,
      created: 12232012,
      status: active,
    }
  ]
}
```

JSON

```
Servers:
-   name: Server1
    owner: John
    created: 12232012
    status: active
```

YAML

Inizio del documento

```
--- # document start  
  
# Comments in YAML look like this.  
  
key: value
```

Chiavi e valori

```
---  
  
key: value  
another_key: Another value goes here.  
a_number_value: 100  
scientific_notation: 1e+12  
boolean: true  
null_value: null  
key with spaces: value
```

Stringhe

```
---
unquoted key: unquoted string value
unquoted key: 'quoted string value'
'Keys can be quoted too.': "Useful if you want to put a ':' in your key."
single quotes: 'have 'one' escape pattern'
double quotes: "have many: \", \0, \t, \u263A, \x0d\x0a == \r\n, and more."
```



Stringhe multilinea

```
---
literal_block: |
    This entire block of text will be the value of the 'literal_block' key,
    with line breaks being preserved.

    The literal continues until de-dented, and the leading indentation is
    stripped.

        Any lines that are 'more-indented' keep the rest of their indentation -
        these lines will be indented by 4 spaces.
folded_style: >
    This entire block of text will be the value of 'folded_style', but this
    time, all newlines will be replaced with a single space.

    Blank lines, like above, are converted to a newline character.

        'More-indented' lines keep their newlines, too -
        this text will appear over two lines.
```

Array di stringhe

Conventional block

```
--- # Favorite movies
- Casablanca
- North by Northwest
- The Man Who Wasn't There
```

Inline format

```
--- # Shopping list
[milk, pumpkin pie, eggs, juice]
```


Oggetti annidati

L'indentazione è necessaria ed ottenuta con spazi (minimo 1, preferibile 2)

```
---  
a_nested_map:  
  key: value  
  another_key: Another Value  
  another_nested_map:  
    hello: hello
```

Array di oggetti

Conventional block

```
---
colors:
- color: blue
  id: 1
- color: red
  id: 2
```

Inline format

```
---
colors: [{color: blue, id: 1},{color: red, id: 2}]
```

Date

```
---  
datetime: 2001-12-15T02:59:43.1Z  
datetime_with_spaces: 2001-12-14 21:59:43.10 -5  
date: 2002-12-14
```

Dati binari

!!binary indica la presenza di un contenuto codificato in Base64

```
---  
gif_file: !!binary |  
R0lGODlhDAAMAIQAAP//9/X17unp5WZmZgAAAOfn515eXvPz7Y6OjuDg4J+fn5  
OTk6enp56enmlpaWNjY6Ojo4SEhP/++f/++f/++f/++f/++f/++f/++f/++f/+  
+f/++f/++f/++f/++f/++SH+Dk1hZGUgd2l0aCBHSU1QACwAAAAADAAMAAFLC  
AgjoEwnuNAFOhpEMTRiggcz4BNJHrv/zCFcLiwMWYNG84BwwEeECcggoBADs=
```

Fine di un documento

```
--- # document start  
key: value  
... # document end  
--- # another document start  
key: value  
... # another document start
```

Recap

```
--- !clarkevans.com/^invoice
invoice: 34843
date : 2001-01-23
bill-to: &id001
  given : Chris
  family : Dumars
  address:
    lines: |
      458 Walkman Dr.
      Suite #292
    city : Royal Oak
    state : MI
    postal : 48046
ship-to: *id001
product:
  - sku : BL394D
    quantity : 4
    description : Basketball
    price : 450.00
  - sku : BL4438H
    quantity : 1
    description : Super Hoop
    price : 2392.00
tax : 251.42
total: 4443.52
comments: >
  Late afternoon is best.
  Backup contact is Nancy
  Billsmer @ 338-4338.
```

SCALAR

COLLECTIONS

MULTI-LINE COLLECTIONS

LISTS/DICTIONARIES

MULTI-LINE FORMATTING

Infine...

YAML è case sensitive

I file devono avere estensione **.yaml** (i **.yml**)

Viene usata l'indentazione con spazi per delineare la struttura (oggetti)

Il carattere TAB non è ammesso

I commenti a singola linea hanno il carattere **#** separato con uno spazio dal testo che segue

Ulteriori informazioni

- Sito ufficiale: <https://yaml.org>
- Ulteriori esempi: <https://learnxinyminutes.com/docs/yaml/>
- Validatore online: <http://www.yamlint.com>